

Patrik Dominik Pördi

+45 (50) 25-8866 | +36 (30) 643-6565 | pordipatrik@gmail.com | Odense, Denmark | [LinkedIn](#) | [Github](#) | [Youtube](#) | [Portfolio](#)

Work Experience

Doctoral researcher

University of Southern Denmark

Odense, Denmark

April-2026 -

- Conducting research in multi-robot systems for agriculture and nature conservation.
- Preparing workshop for ROSCon 2026, with multi robot system additions.

Robotics & Drone Engineer

ARK Electronics

Salt Lake City, United States

June-2024 to March 2026

- Delivered ROS2 tutorials on custom PX4 flight modes & Computer Vision applications, presented and held a workshop at ROSCon 2024 and 2025, respectively.
- Engineered GPS/IMU drivers, integrated PX4 on NVIDIA Jetson and Raspberry Pi platforms, and implemented encrypted logging for secure flight data.
- Performed systematic flight testing, contributed to drone design and diagnostics, and provided mentorship and technical guidance to interns.
- Brought extensive hardware expertise, including soldering, board-level troubleshooting, and seamless integration of third-party modules across embedded and autonomous systems.

Teaching Assistant in Robot Programming Classes

University of Maryland

College Park, United States

August-2023 to June-2024

- Delivered Python/C++ ROS 2 labs, held office hours, supported course setup and grading, supervised a TurtleBot maze competition for both undergraduate and graduate students.

UAV Engineer

Aviant AS

Trondheim, Norway

June-2023 to August-2023

- Performed big data analytics on Ulog data to evaluate control performance and measurement accuracy, while developing a computer vision-based system to track package deflection during winching.
- Enabled precise delivery in densely populated areas by implementing a high-accuracy RTK GPS solution.

Electrical Engineer

Evosoft Hungaria Kft. member of Siemens AG

Budapest, Hungary

July-2020 to August-2021

- Designed a CNC cyber-physical system via OPC UA.
- Developed test environment and handled PLC programming of Siemens drive systems using TIA Portal.

Education

University of Maryland, College Park

Master of Engineering in Robotics - GPA: 3.88/4.0

College Park, Maryland, May 2024

- Specialization in Autonomous Robotics
- Projects: Stochastic control of Drones, Geometric decentralized Drone swarms, 3D Vision and mapping

Budapest University of Technology and Economics | BME

Master of Mechatronics Engineering - GPA: 4.51/5.0

Budapest, Hungary, May 2023

- Specialization in Cyber-Physical Systems
- Thesis: Design and implementation RMS and Industry 4.0 into a CNC manufacturing system
- Exchange semester online at [University Technology of Sydney](#)
- Exchange year at [Arctic University of Norway](#)

Sydney, Australia, June 2022

Narvik, Norway, June 2022

Bachelor of Mechatronics Engineering - GPA: 4.48/5.0

Budapest, Hungary, January 2021

- Specialization in Cyber-Physical Systems
- Thesis: Implementation of CNC Lathe Digital Twin simulation using Siemens tools
- German bilingual program: exchange at [Karlsruhe Institute of Technology](#)

Karlsruhe, Germany, February 2020

Skills

Languages: English | non-native fluent, German | non-native fluent, Hungarian | native

Software: C, C++, ROS2, Python, Rust, PX4, ArduPilot, Betaflight, MATLAB & Simulink, Linux CLI, Git, OpenCV, PLC, PyTorch3D

Modeling: Gazebo, Foxglove, PlotJuggler, SolidWorks, Inkscape, Siemens Software, Visual Components, 3D Printing

Expertise: Autonomous Robotics, Data Science & Physical AI, Digital Twins, Path Planning, Machine Learning, Computer & 3D Vision, SLAM

Research experience & activities

Dronecode Foundation

PX4 Autopilot ROS2 maintainer

United States
October 2025-

- Maintaining the ROS2 support on PX4, implemented Zenoh middleware

HYPE (Hungarian Young Professional Engagement) Network

Hungarian & American Leadership Network

Washington DC, United States
May 2025-

- Participated in monthly seminars on effective leadership, delivered by prominent American leaders

IEEE RAS

P2817- Verification of Autonomous Systems

United States
May 2024-

- Participating on meetings to work on Robotics Standards

Competing in NIST 2023 First Responder UAS 3D Mapping Challenge

Autonomous micro air vehicle (AMAV) team, UMD

Maryland, United States
September 2023-

- Implementing 3D mapping solutions on the drone.

Research on STEP file-based automatic production time estimation

Department of Manufacturing Science and Engineering, BME

Budapest, Hungary
January 2020-June 2022

- A software application was created using Python and regular expressions (RegEx) that can extract machining features from any provided STEP file and then estimating production-related measurements.

Research on Reconfigurable Manufacturing Systems with Robotics concentration

Department of Industrial Engineering, UiT

Narvik, Norway
January 2022-June 2023

- Multiple versions of RMS were developed for oil industry scenarios using Python and Visual Components.

Engineering student project team member

BME Mechatronics Student Society

Budapest, Hungary
January 2018-June 2022

- Investigated software and hardware development for projects, such as line-follower mini car, jumping robot, closed chain mechanism with inverse kinematics simulation.
- Held consultations in programming, manufacturing, and robot modeling subjects.

Scholarships & Honors

Rosztoczy Foundation Scholarship Program - Hungarian Scholarship	2023
Hungarian American Coalition Dr. Elemér and Éva Kiss Scholarship	2022, 2023
Second Place, Hungarian National Round of the Conference of Student Research	2023
Presented a paper on STEP file-based automatic production time estimation	
Fulbright Scholar Award for Master's Studies in the United States	2022
First Place, Conference of Student Research at BME (Budapest University of Technology and Economics)	2021
Qualified for the National Student Research Conference with a paper on STEP file-based automatic production time estimation	
Professional Scholarship for Exceptional Members of the BME German Bilingual Class	2019
BME Study & Sport Scholarship	2020
Outstanding Achievements in Hungarian Handball Championships	2020
Third Place, Hungarian University - College National Handball Championship	2020